

# Experiment No. 8: Customer Churn Prediction using Decision Tree Classification

## Aim

To build and evaluate a Decision Tree Classification model for predicting customer churn using a real-world dataset and analyze the factors influencing customer retention.

## Questions & Answers

### Q1. What is classification, and how does it differ from regression?

Answer:

- **Classification:** A supervised learning task that maps input features to categorical class labels (e.g., *Churn* vs. *No Churn*).
- **Regression:** A supervised learning task that maps input features to a continuous numerical target (e.g., *Predicting House Prices*).

| Aspect                     | Classification                        | Regression                    |
|----------------------------|---------------------------------------|-------------------------------|
| Output Type                | Discrete / Categorical labels         | Continuous numerical values   |
| Example                    | Email Spam vs. Not Spam               | Predicting future stock price |
| Primary Evaluation Metrics | Accuracy, Precision, Recall, F1-Score | MAE, MSE, RMSE, R^2 Score     |

### Q2. Explain the working principle of the Decision Tree algorithm.

Answer:

A Decision Tree builds a flowchart-like tree structure by recursively partitioning the training dataset into smaller subsets:

- **Root Node Selection:** Evaluates candidate feature split points using impurity criteria (Gini Index or Entropy) to select the optimal root node feature.
- **Splitting Phase:** Splits data into branches based on feature threshold conditions.
- **Recursive Partitioning:** Repeats the splitting process on child sub-nodes.
- **Terminal / Leaf Nodes:** Stops splitting upon hitting predefined stopping criteria (e.g., `max_depth`, `min_samples_split`) and assigns class labels via majority voting.

### Q3. Differentiate between Gini Index and Entropy as splitting criteria.

Answer:

| Attribute           | Gini Index                                    | Entropy                                              |
|---------------------|-----------------------------------------------|------------------------------------------------------|
| Formula             | $1 - \sum_{i=1}^C (p_i)^2$                    | $-\sum_{i=1}^C p_i \log_2(p_i)$                      |
| Range               | 0 to 0.5 (for binary classification)          | 0 to 1 (for binary classification)                   |
| Computational Speed | Faster (no logarithmic calculations required) | Slower (requires log calculations)                   |
| Default Usage       | Default impurity measure in Scikit-learn      | Preferred when measuring Information Gain explicitly |

#### **Q4. What is a Confusion Matrix? Explain its components.**

**Answer:**

A **Confusion Matrix** is a  $2 \times 2$  performance evaluation table comparing actual target classes with predicted classes:

- **True Positive (TP):** Model correctly predicted a churning customer as *Churn*.
- **True Negative (TN):** Model correctly predicted a retained customer as *No Churn*.
- **False Positive (FP - Type I Error):** Model incorrectly flagged a retained customer as *Churn*.
- **False Negative (FN - Type II Error):** Model incorrectly flagged an actual churning customer as *No Churn*.

#### **Q5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?**

**Answer:**

- **Accuracy:** Overall proportion of correct predictions across all classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Ratio of correctly predicted positive observations to total predicted positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** Ratio of correctly predicted positive observations to all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** Harmonic mean of Precision and Recall, crucial for handling imbalanced datasets.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

#### **Q6. What is overfitting in a Decision Tree? How can it be reduced?**

**Answer:**

- **Overfitting:** Occurs when a Decision Tree grows unconstrained, memorizing noisy training data details and creating overly complex decision boundaries. This results in high training accuracy but poor generalization on test data.
- **Reduction Strategies:**
  - **Pre-Pruning:** Setting hyperparameter limits like `max_depth`, `min_samples_split`, or `min_samples_leaf`.
  - **Post-Pruning:** Trimming branches after full tree growth using Cost Complexity Pruning (`ccp_alpha`).
  - **Ensemble Methods:** Using Random Forests or Gradient Boosted Trees.

#### **Q7. Why is customer churn prediction important for businesses?**

**Answer:**

- **Cost Efficiency:** Acquiring a new customer is up to 5x to 25x more expensive than retaining an existing customer.
- **Revenue Protection:** Prevents recurring revenue losses from high-margin subscription accounts.
- **Proactive Interventions:** Allows marketing teams to issue targeted retention offers before customers formally leave.

#### **Q8. What are the advantages and limitations of the Decision Tree algorithm?**

**Answer:**

- **Advantages:**
  - High interpretability; decision rules can be easily visualized and explained.

- Requires minimal data preprocessing (handles numerical and categorical features naturally).
- Non-parametric; makes no strict assumptions about underlying data distributions.
- **Limitations:**
  - Highly prone to overfitting if not properly pruned.
  - Unstable; small changes in input data can yield vastly different tree structures.
  - Can struggle with complex non-linear relationships compared to gradient boosting.

**Q9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.**

**Answer:**

- **Banking & Credit Scoring:** Evaluating loan applicants as *High Risk* or *Low Risk*.
- **Healthcare & Diagnosis:** Classifying medical test results to detect specific diseases.
- **Cybersecurity:** Classifying incoming network traffic as *Legitimate* or *Malicious Intrusion*.

**Q10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?**

**Answer:**

- **Targeted Incentives:** Identifies high-churn risk segments to send customized discount vouchers or loyalty extensions rather than blanket promotions.
- **Product Enhancements:** Reveals key churn drivers (e.g., month-to-month contracts or lack of tech support), allowing business teams to optimize product tiers and improve user onboarding.
- **Maximizing Customer Lifetime Value (CLV):** Extends average account lifespan, driving sustainable profit margins.